Cover letter

Head of Unit E2
European Commission
Health and Food Safety Directorate-General
Directorate E - Safety of the food chain
Unit E2 - Food Processing Technologies and Novel Foods
B-1049 Brussels

Date: Feb 18 2025

Subject: Application for authorisation of food enzyme in accordance with Regulation (EC) No 1331/2008

☐ Application for an authorisation of a new food enzyme
☒ Application for a modification of the conditions of use of an already authorised food enzyme
☐ Application for a modification of the specifications of an already authorised food enzyme

The Applicant:

Acme Enzymatic srl
Via del Prototipo 18
00144 Roma
Italia

submits the present application in order to update the EU list on food enzymes.

Food enzyme name (Accepted Name according to the IUBMB):

Alpha-amylase

Enzyme Classification Number of Enzyme Commission of the IUBMB:

EC 3.2.1.1

Source material:

A genetically modified strain of *Bacillus licheniformis* (strain NZYM-KE)

Acme Enzymatic srl proposes that the above food enzyme is entered into the future EU Community list of food enzymes as indicated below, according to the format foreseen by Article 7.2 of Regulation (EC) No 1332/2008:

Name	Specifications	Foods	Conditions of use	Restrictions on the sale of the food enzyme directly to the final consumer	Specific requirements in respect of the labelling of food
Alpha-amylase	**IUBMB no.:** EC 3.2.1.1 **Systematic name:** 1,4-alpha D-glucan glucanohydrolase **Other names:** Alpha-amylase **Source:** A genetically modified strain of *Bacillus licheniformis* **Strain designation:** NZYM-KE **Identification:** Alpha-amylases catalyse the hydrolysis of 1,4-alpha-glycosidic linkages in the starch polysaccharides amylose and amylopectin **Purity criteria:** All preparations placed on the market shall comply with the general purity specifications set by JECFA (2006) for food enzymes	***In present dossier:*** Production of: - baked products - cereal-based products other than baked - plant-based analogues of milk and milk products - starch and gluten fractions - refined and unrefined sugar ***In original dossier:*** Production of: - distilled alcohol - glucose syrups and other starch hydrolysates	*quantum satis*	Not intended for direct sale to final consumer	None - the food enzyme does not perform a technological function in final foods nor can its use mislead the consumers.

Applicant contact and signatory

Document reference: AE-REG-2025-018
Date: 18 February 2025

For and on behalf of Acme Enzymatic srl
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Typed signatory: Elena Marinelli
Synthetic contact details; no legal signature.

Enclosures:

☒ Complete dossier
☒ Public summary of the dossier
☒ Detailed summary of the dossier
☐ List of parts of the dossier requested to be treated as confidential accompanied by verifiable justification demonstrating how the disclosure of such information would potentially harm the interests of the application to a significant degree
☐ List of studies and all information concerning the notification of the studies in accordance with Article 32b of Regulation (EC) No 178/2002
☒ Checklist for completeness

1. Public summary

Scope of the application

The present dossier is an application for extension of use of the food enzyme alpha-amylase produced by a strain of *Bacillus licheniformis* (strain NZYM-KE) which was previously submitted (EFSA-Q-2012-00898) and concluded by EFSA with a positive opinion. The original submitted dossier is referred to as “original dossier" hereafter.

As documented in the present dossier the extension of use of the food enzyme complies with the criteria foreseen by Regulation (EC) 1332/2008 on food enzymes, i.e.

- the food enzyme is safe for consumers in its intended uses
- it fulfils a reasonable technological need in food processing
- its use does not mislead consumers.

Identity of the food enzyme

Name of the enzyme: Alpha-amylase.

IUBMB number: EC 3.2.1.1.

Source material and production method

The source of the enzyme remains the same as described in the original dossier, *Bacillus licheniformis* (strain NZYM-KE). The manufacturing process for the enzyme also remains consistent with the original dossier.

Toxicological studies

The food enzyme production strain qualifies for the QPS approach of safety assessment, and no concerns arise from the product and the production process. Consequently, no toxicological testing other than assessment of allergenicity is necessary for the safety assessment of the food enzyme. The assessment of allergenicity submitted in the original dossier remain applicable and continue to provide valid safety data for the enzyme.

Uses of the food enzyme in food production

The food enzyme can be used in various food manufacturing processes. In these processes the food enzyme is added during the food production process where it performs its function. The food enzyme has no function in the final food because the enzyme protein is denatured by high temperature or removed during the food manufacturing process, or because there is no substrate for the enzyme.

Technological need of the food enzyme

Alpha-amylases have been used in food manufacturing processes for many years, which by itself demonstrates their technological need.

The use of the food enzyme will not mislead consumers

The food enzyme is not functional in the final food, and in many cases the use of the enzyme does not result in effects on the final food.

In cases where there may be a noticeable effect on the final food it is likely that the same effect could be achieved by a similar process without the use of the food enzyme.

In any case, the final food products or their characteristics are not different from what is expected by the consumer based on appearance or already available information from the product label.

Based on this there is no reason to believe that the use of the food enzyme could be misleading for the consumer.

Conclusions on the safety of the food enzyme

It is concluded that the use of the food enzyme in food processing is safe. This is based on the safety evaluation of the food enzyme, the production microorganism, the enzyme manufacturing process and its intended uses in food processing.

2. Detailed summary

The present dossier is an application for an extension of use of the food enzyme alpha-amylase produced by a strain of *Bacillus licheniformis* (strain NZYM-KE). An application for authorisation of this food enzyme was previously submitted (EFSA-Q-2012-00898) and concluded by EFSA with a positive opinion[1] published in March 2021.

The original submitted dossier is referred to as “original dossier” hereafter.

Identity of the food enzyme

Accepted IUBMB name and number of the food enzyme: Alpha-amylase, EC 3.2.1.1.

The food enzyme complies with the internationally accepted JECFA specifications for chemical and microbiological purity of food enzymes, i.e.:

Lead:	Not more than 5 mg/kg
Salmonella sp.:	Absent in 25 g of sample
Total coliforms:	Not more than 30 per gram
Escherichia coli:	Absent in 25 g of sample
Antimicrobial activity:	Not detected

Source material and production method

The source of the food enzyme remains the same as described in the original dossier, *Bacillus licheniformis* (strain NZYM-KE). The food enzyme manufacturing process described in the original dossier still applies.

Toxicological studies

The food enzyme production strain qualifies for the QPS approach of safety assessment, and no concerns arise from the product and the production process. Consequently, no toxicological testing other than assessment of allergenicity is necessary for the safety assessment of the food enzyme. The assessment of allergenicity submitted in the original dossier (EFSA-Q-2012-00898) remains applicable and continues to provide valid safety data for the enzyme.

Reaction and fate in food

The food enzyme catalyses the hydrolysis of amylose and amylopectin into soluble dextrins and other maltooligosaccharides.

[1] Safety evaluation of the food enzyme α-amylase from the genetically modified Bacillus licheniformis strain NZYM-KE

The substrates and the reaction products are themselves present in food ingredients. Consequently, no adverse effect on nutrients is expected.

The reaction products from the hydrolysis reaction are soluble dextrins and other malto-oligosaccharides. Such molecules are naturally present in cereal-containing foods, which are a part of the ordinary consumer's diet. Therefore, they do not create any safety concern.

The food enzyme object of the present dossier only catalyses the above-described reaction, therefore there is no unintended reaction product.

The food enzyme performs its technological function during food processing and does not perform any technological function in the final food since the enzyme is denatured or removed.

Fate of this food enzyme during food processes is described in detail in its dedicated sections.

Based on the conditions of use and the activity of the enzyme under such conditions it can be concluded that the food enzyme does not exert any (unintentional) enzymatic activity in the final food.

Proposed uses in food processing

In principle, the enzymatic conversion of starch with the help of the alpha-amylase can be used in the processing of all food raw materials which naturally contain starch.

The **current dossier** presents the extension of the use of the food enzyme in five additional food manufacturing processes:

- Production of baked products
- Production of cereal-based products other than baked
- Production of plant-based analogues of milk and milk products
- Production of starch and gluten fractions
- Production of refined and unrefined sugar

The use of the food enzyme in these food manufacturing processes are summarised further below.

The **original dossier** (EFSA-Q-2012-00898) describes the intended use of alpha-amylase in the following two food manufacturing processes:

- Production of distilled alcohol
- Production of glucose syrups and other starch hydrolysates

Production of baked products

In the production of baked products food enzyme are used to hydrolyse starch and release dextrins that are further degraded into fermentable sugars. This process improves dough structure and extensibility, and enhances gas production during fermentation.

Production of cereal-based products other than baked

In the production of cereal-based products other than baked food enzyme is added to the liquefied slurry to reduce viscosity, allowing for smooth processing and increases the inclusion of cereals in food and beverages.

Production of plant-based analogues of milk and milk products

In the production of plant-based analogues of milk and milk products the food enzyme is added to the liquefied slurry to hydrolyse gelatinised starch and thereby reduce viscosity allowing for increased inclusion of cereals in the product and ensure smooth processing.

Production of starch and gluten fractions

In the production of starch and gluten fractions the food enzyme can be added in the solubles fraction during evaporation to remove residual starch in this fraction to improve processability, as retrogradated starch otherwise makes a strong gel and increases viscosity which negative impact processability.

Production of refined and unrefined sugar

In the production of refined and unrefined sugar the food enzyme can be used to remove starch. High starch content in raw sugar can cause filtration issues, poor crystallization, and reduced yield. It may also create a visible haze in refined sugar, which is undesirable in products like soft drinks. Using alpha-amylase at the start of the process helps prevent these problems.

Dietary exposure

As described in the original dossier and in the current use extension dossier the food enzyme is intended for use in a total of seven food manufacturing processes. In two of these food manufacturing processes (production of distilled alcohol, and glucose syrups and other starch hydrolysates) the enzyme and Total Organic Solids (TOS) are removed during certain steps of the food manufacturing processes and consequently is absent in the final food. This is also the case for use of the food enzyme in production of starch and gluten fractions. The food enzyme is used in the evaporation step of the solubles fraction. This fraction is used only for animal feed or as starting material for production of distilled alcohol, and since distillation removes all enzyme TOS, no TOS is present in the final alcohol. Accordingly, no dietary exposure is calculated for use of the enzyme in these three processes. For use of the enzyme in the remaining four food manufacturing processes, an estimated dietary exposure of the enzyme is calculated by using the FEIM web-tool calculator.

The calculated estimated dietary exposure was found to be 0.71 mg Total Organic Solids (TOS) / kg body weight (bw) / day for 'toddlers'.

It must be emphasised that this intake calculation is based on conservative assumptions and represents a highly exaggerated value because of the following reasons:

- It is assumed that all producers of the above mentioned foodstuffs use the specific food enzyme object of this application
- It is assumed that all producers apply the highest use level per application
- It is assumed that the amount of TOS does not decrease as a result of the food production process
- It is assumed that the final food containing the calculated theoretical amount of TOS is consumed daily over the course of a lifetime

The effect of the food enzyme on the final food

Effect of the presence of (residues of) the food enzyme on the final food

It is not the food enzyme itself but the result of the enzymatic conversion that determines the effect in the food or food ingredient (including raw materials). This effect remains, irrespective of whether the food enzyme is still present or removed from the final food.

The alpha-amylase is used for hydrolysis of amylose and amylopectin during processing of starch-containing foods. It is used in food manufacturing as a processing aid to ease and optimise the processes and make more efficient use of raw materials. Also, the use of alpha-amylase ensures a consistent process and minimises potential end product variations.

The enzyme does not exert any (unintentional) enzymatic activity in the final food, due to a combination of various factors, depending on the application and the process conditions used by the individual food producer. These factors include depletion of the substrate, denaturation of the enzyme during processing, lack of water activity, wrong pH, etc.

Consequently, the presence of (residues of) the food enzyme in the final food does not lead to an effect in or on the final food.

Effect of the use of the food enzyme on the final food

The majority of food enzymes perform their technological function during food processing. In many cases, they are used merely to improve the production process of a certain food or food ingredient, and the effect of the enzymatic conversion is not noticeable in the final food. In other cases, however, the effect of the enzymatic conversion (depletion of a substrate and/or development of a reaction product) might also result in effects on the final food.

The benefits that are described above when describing the technological need of the food enzyme are, however, not exclusively obtainable by means of enzyme treatment but can be achieved without the use of enzymes through *e.g.*, modified, maybe more expensive or less environmentally friendly production processes or recipe changes.

Why the use would not mislead the consumer

The effect of the enzymatic conversion with the help of the food enzyme is the degradation or modification of the substrate, resulting in improved processing properties during production of various food products.

The food enzyme does not perform any technological function in the final foods containing ingredients prepared with the help of this enzyme. Moreover, the food products prepared with the help of alpha-amylase do not have other characteristics than what is expected by the consumer.

Alpha-amylases have been used for the above-described technological function in food processes for many years and have been specifically approved in many countries including Denmark and France.

Considering the above there are no reasons to believe that the use of the food enzyme in food processing could be misleading for the consumer.

Overall conclusions on the safety of the food enzyme

It is concluded that the use of the alpha-amylase produced by *Bacillus licheniformis* strain NZYM-KE in food processing is safe. This is based on safety evaluation of the food enzyme, the production microorganism, the food enzyme manufacturing process and its intended use in food processing.

Consequently, it is concluded that the alpha-amylase produced by *Bacillus licheniformis* strain NZYM-KE can be safely used *quantum satis* in the intended food manufacturing processes, and even in food processing in general.

10. Reaction and fate in food to which the food enzyme is added

The enzyme and its substrate

The principal enzyme activity is alpha-amylase. Alpha-amylases are widespread in nature and are universally distributed throughout the animal, plant (e.g. wheat and barley) and microbial kingdom (bacteria and fungi). Alpha-amylases are naturally present in many raw materials including wheat, barley and malt.

Alpha-amylases catalyse the hydrolysis of 1,4-alpha-glycosidic linkages in the starch polysaccharides amylose and amylopectin, which are found in various plant material including cereals, such as wheat and barley. The enzymatic hydrolysis of amylose and amylopectin results in the generation of soluble dextrins and other maltooligosaccharides. Like the substrate and the enzyme, these oligosaccharides are natural constituents of cereal-containing foods, which are a part of the ordinary consumer's diet. Therefore, they do not create any safety concern.

The food enzyme object of the present dossier only catalyses the above-described reactions; therefore, there is no unintended reaction product.

Fate of the enzyme when used in food processing

The properties of the food enzyme are technologically beneficial in various industrial food manufacturing processes.

The food enzyme is used as a processing aid in various food manufacturing processes.

These food manufacturing processes include processing steps where the food enzyme is denatured or removed, and therefore the food enzyme does not perform any technological function in the final food.

The enzyme protein itself is similar to other proteins in food and is not expected to have any significant interaction with food components. The starch hydrolysis products, which are described above, are normal constituents of the human diet.

In conclusion, the food enzyme does not exert any (unintentional) enzymatic activity, and it does not create any safety-related undesirable effects during its use in food processing.

11. Intended use(s) in food and use level(s) (Proposed normal and maximum use levels)

Further details regarding the information presented in this section can be found in the 'Risk management' part of this dossier.

The food enzyme is used as a processing aid in food manufacturing processes and is not added directly to final foodstuffs.

The food enzyme is intended for use in a total of seven food manufacturing processes:

The **current dossier** presents the extension of the use of the food enzyme in five additional food manufacturing processes:

- Production of baked products
- Production of cereal-based products other than baked
- Production of plant-based analogues of milk and milk products
- Production of starch and gluten fractions
- Production of refined and unrefined sugar

The **original dossier** (EFSA-Q-2012-00898) describes the intended use of alpha-amylase in the following two food manufacturing processes:

- Production of distilled alcohol
- Production of glucose syrups and other starch hydrolysates

In accordance with the principles of Good Manufacturing Practice (GMP) the food enzyme is used at the minimum dosage necessary to achieve the desired enzymatic reaction.

The use level applied in practice by a food manufacturer depends on the particular food manufacturing process. It is based on an initial recommendation by the enzyme manufacturer and optimised to fit the process conditions.

The recommended use levels for use of the food enzyme in each food manufacturing process and the physical raw material to which the food enzyme is added during food processing.

Food manufacturing process Production of:	**Raw material (RM)**	**Maximum recommended use levels** (mg TOS per kg RM)	
		Current dossier	***Original dossier***
distilled alcohol	fermentable carbohydrates	32.7	32.7
glucose syrups and other starch hydrolysates	starch	32.7	32.7
baked products	flour	50	-
cereal-based products other than baked	flour	50	-
plant-based analogues of milk and milk products	plant-based raw materials	121	-
starch and gluten fractions	cereals	1.6	-
refined and unrefined sugar	sugar beets and sugar canes	1.6	-

12. Dietary exposure assessment

Overall, the human exposure to the food enzyme will be negligible because the enzyme preparation is used as a processing aid and in low dosages.

The food enzyme is intended for use in a total of seven food manufacturing processes:

The **current dossier** presents the extension of the use of the food enzyme in five additional food manufacturing processes:

- Production of baked products
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- Production of distilled alcohol
- Production of glucose syrups and other starch hydrolysates

In two of these food manufacturing processes (production of distilled alcohol, and glucose syrups and other starch hydrolysates) the enzyme and Total Organic Solids (TOS) are removed during certain steps of the food manufacturing processes and consequently is absent in the final food. This is also the case for use of the food enzyme in production of starch and gluten fractions. The food enzyme is used in the evaporation step of the solubles fraction. This fraction is used only for animal feed or as starting material for production of distilled alcohol, and since distillation removes all enzyme TOS, no TOS is present in the final alcohol. Accordingly, no dietary exposure is calculated for use of the enzyme in these three processes.

For use of the enzyme in the remaining four food manufacturing processes, an estimated dietary exposure of the enzyme is calculated by using the FEIM web-tool[1], applying the standard, default parameters in the tool.

This intake calculations represent a “maximum worst case” situation of human consumption for the following reasons:

- It is assumed that all producers of the specific foodstuffs use the food enzyme in their production processes
- It is assumed that the food enzyme is used at its maximum recommended dosages
- It is assumed that the amount of TOS does not decrease as a result of the food production process

[1] Feim - Default (europa.eu)

- It is assumed that the final food containing the calculated theoretical amount of TOS is consumed daily over the course of a lifetime

Therefore, the estimated exposure calculated with this method is highly conservative.

Applying the maximum recommended use level of the food enzyme in each of above mentioned four food processes in the FEIM web-tool, the following exposure estimates for infants, toddlers, children, adolescents, adults and the elderly, are found.

POPCLASS	N mean mg TOS / kg BW / day	MIN mean mg TOS / kg BW / day	MAX mean mg TOS / kg BW / day	N 95th P mg TOS / kg BW / day	MIN 95th P mg TOS / kg BW / day	MAX 95th P mg TOS / kg BW / day
All	All	All	All	All	All	All
Infants	12	0.048	0.27	11	0.14	0.68
Toddlers	15	0.14	0.39	14	0.32	0.71
Other children	19	0.079	0.33	19	0.19	0.61
Adolescents	21	0.023	0.2	20	0.059	0.35
Adults	22	0.062	0.12	22	0.13	0.27
Elderly and very elderly	23	0.057	0.12	22	0.12	0.23

Accordingly, the population group with the highest estimated accumulated dietary exposure is ‘toddlers’ with an estimated total maximum daily intake of 0.71 mg TOS/kg bw/day.

Margin of exposure

No margin of exposure is calculated, since no toxicological assessment is considered necessary, because the food enzyme meets all criteria for the QPS approach of safety assessment (EFSA CEP Panel, 2021).

13. Overall conclusion on the safety of the food enzyme

In conclusion, the alpha-amylase produced by *Bacillus licheniformis* NZYM-KE is safe for its intended use in food processing.

This is based on the safety evaluation of the food enzyme, the production microorganism, the enzyme manufacturing process, and its intended use in food processing as described in the original dossier (EFSA-Q-2012-00898, evaluated by EFSA[1]) and the use extension described in the present dossier.

[1] Safety evaluation of the food enzyme α-amylase from the genetically modified Bacillus licheniformis strain NZYM-KE

11. Intended use(s) in food and use level(s) (Proposed normal and maximum use levels)

Further details regarding the information presented in this section can be found in the 'Risk management' part of this dossier.

The food enzyme is used as a processing aid in food manufacturing processes and is not added directly to final foodstuffs.

The food enzyme is intended for use in a total of seven food manufacturing processes:

The **current dossier** presents the extension of the use of the food enzyme in five additional food manufacturing processes:

- Production of baked products
- Production of cereal-based products other than baked
- Production of plant-based analogues of milk and milk products
- Production of starch and gluten fractions
- Production of refined and unrefined sugar

The **original dossier** (EFSA-Q-2012-00898) describes the intended use of alpha-amylase in the following two food manufacturing processes:

- Production of distilled alcohol
- Production of glucose syrups and other starch hydrolysates

In accordance with the principles of Good Manufacturing Practice (GMP) the food enzyme is used at the minimum dosage necessary to achieve the desired enzymatic reaction.

The use level applied in practice by a food manufacturer depends on the particular food manufacturing process. It is based on an initial recommendation by the enzyme manufacturer and optimised to fit the process conditions.

The recommended use levels for use of the food enzyme in each food manufacturing process and the physical raw material to which the food enzyme is added during food processing.

Food manufacturing process Production of:	**Raw material (RM)**	**Maximum recommended use levels** (mg TOS per kg RM)	
		Current dossier	***Original dossier***
distilled alcohol	fermentable carbohydrates	32.7	32.7
glucose syrups and other starch hydrolysates	starch	32.7	32.7
baked products	flour	50	-
cereal-based products other than baked	flour	50	-
plant-based analogues of milk and milk products	plant-based raw materials	121	-
starch and gluten fractions	cereals	1.6	-
refined and unrefined sugar	sugar beets and sugar canes	1.6	-

12. Dietary exposure assessment

Overall, the human exposure to the food enzyme will be negligible because the enzyme preparation is used as a processing aid and in low dosages.

The food enzyme is intended for use in a total of seven food manufacturing processes:

The **current dossier** presents the extension of the use of the food enzyme in five additional food manufacturing processes:

- Production of baked products
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The **original dossier** (EFSA-Q-2012-00898) describes the intended use of alpha-amylase in the following two food manufacturing processes:

- Production of distilled alcohol
- Production of glucose syrups and other starch hydrolysates

In two of these food manufacturing processes (production of distilled alcohol, and glucose syrups and other starch hydrolysates) the enzyme and Total Organic Solids (TOS) are removed during certain steps of the food manufacturing processes and consequently is absent in the final food. This is also the case for use of the food enzyme in production of starch and gluten fractions. The food enzyme is used in the evaporation step of the solubles fraction. This fraction is used only for animal feed or as starting material for production of distilled alcohol, and since distillation removes all enzyme TOS, no TOS is present in the final alcohol. Accordingly, no dietary exposure is calculated for use of the enzyme in these three processes.

For use of the enzyme in the remaining four food manufacturing processes, an estimated dietary exposure of the enzyme is calculated by using the FEIM web-tool[1], applying the standard, default parameters in the tool.

This intake calculations represent a “maximum worst case” situation of human consumption for the following reasons:

- It is assumed that all producers of the specific foodstuffs use the food enzyme in their production processes
- It is assumed that the food enzyme is used at its maximum recommended dosages
- It is assumed that the amount of TOS does not decrease as a result of the food production process

[1] Feim - Default (europa.eu)

- It is assumed that the final food containing the calculated theoretical amount of TOS is consumed daily over the course of a lifetime

Therefore, the estimated exposure calculated with this method is highly conservative.

Applying the maximum recommended use level of the food enzyme in each of above mentioned four food processes in the FEIM web-tool, the following exposure estimates for infants, toddlers, children, adolescents, adults and the elderly, are found.

POPCLASS	N mean mg TOS / kg BW / day	MIN mean mg TOS / kg BW / day	MAX mean mg TOS / kg BW / day	N 95th P mg TOS / kg BW / day	MIN 95th P mg TOS / kg BW / day	MAX 95th P mg TOS / kg BW / day
All	All	All	All	All	All	All
Infants	12	0.048	0.27	11	0.14	0.68
Toddlers	15	0.14	0.39	14	0.32	0.71
Other children	19	0.079	0.33	19	0.19	0.61
Adolescents	21	0.023	0.2	20	0.059	0.35
Adults	22	0.062	0.12	22	0.13	0.27
Elderly and very elderly	23	0.057	0.12	22	0.12	0.23

Accordingly, the population group with the highest estimated accumulated dietary exposure is ‘toddlers’ with an estimated total maximum daily intake of 0.71 mg TOS/kg bw/day.

Margin of exposure

No margin of exposure is calculated, since no toxicological assessment is considered necessary, because the food enzyme meets all criteria for the QPS approach of safety assessment (EFSA CEP Panel, 2021).

Risk Management

14. The function and technological need, including a description of the typical process(es) in which the food enzyme may be applied

The enzyme and its substrate

The principal enzyme activity is alpha-amylase. Alpha-amylases are widespread in nature and are universally distributed throughout the animal, plant (e.g. wheat and barley) and microbial kingdom (bacteria and fungi). Alpha-amylases are naturally present in many raw materials including wheat, barley and malt.

Like any other enzyme the food enzyme object of the present dossier acts as a biocatalyst. With the help of the enzyme a certain substrate is converted into a certain reaction product. As the catalytic activity is very specific, it is not to be expected that the food enzyme will act in another way than described below.

After complete conversion the enzyme is no longer functioning.

Alpha-amylases catalyse the hydrolysis of 1,4-alpha-glycosidic linkages in the starch polysaccharides amylose and amylopectin, which are found in various plant material including cereals, such as wheat and barley. The enzymatic hydrolysis of amylose and amylopectin results in the generation of soluble dextrins and other maltooligosaccharides. Like the substrate and the enzyme, the reaction products are natural constituents of cereal-containing foods.

Use of the enzyme in food processing

The properties of the food enzyme are technologically beneficial in various industrial food manufacturing processes, and the reaction products, which are described above, are normal constituents of the human diet. Alpha-amylase is naturally present in many raw materials.

In principle, alpha-amylases can be used in the processing of all food raw materials, which naturally contain the substrates.

Since the 1960s alpha-amylases have been used extensively in various industrial food applications for the hydrolysis of starch, with its major application in the starch, brewing, distilling and baking industry (Reed, 1966, Pence, 1953, Beg et al, 2001). Some of the technical aspects of the use of alpha-amylase in sweetener production, brewing, alcohol distillation, baking and fruit juice production are described in the book ‘Enzyme arbeiten für uns’ (Uhlig, 1991).

The food enzyme object of the present dossier is intended for use as a processing aid in a total of seven food manufacturing processes as described elsewhere in this dossier.

The **current dossier** presents the extension of the use of the food enzyme in five additional food manufacturing processes:

- Production of baked products
- Production of cereal-based products other than baked
- Production of plant-based analogues of milk and milk products
- Production of starch and gluten fractions
- Production of refined and unrefined sugar

The **original dossier** (EFSA-Q-2012-00898) describes the intended use of alpha-amylase in the following two food manufacturing processes:

- Production of distilled alcohol
- Production of glucose syrups and other starch hydrolysates

Production of baked products

Alpha-amylases can be used in the production of baked products such as, but not limited to, bread, biscuits, steamed bread, cakes, pancakes, tortillas, wafers and waffles.

Starch constitutes the major component of wheat flour, 74-90% on a dry basis depending on the extraction rate from grain milling. In baking, starch provides the yeast with fermentable sugars and it contributes to the structure of the crumb and to the structure and colour of the crust. Starch-protein interactions are important during all stages of the baking process.

During production of baked products, alpha-amylase hydrolyses the starch from granules that have been damaged during the milling process and hereby releases dextrins. The amount of damaged starch is dependent on wheat variety and milling process (typically 7-9% of the starch level). The resulting dextrins are further degraded by the cereal beta-amylase or an additional glucoamylase generating sugars which can be fermented by yeast.

The level of alpha-amylase varies in cereals depending on cereal type, growing and harvesting conditions, and therefore adjustment of alpha-amylase levels is important to ensure a consistent process and product.

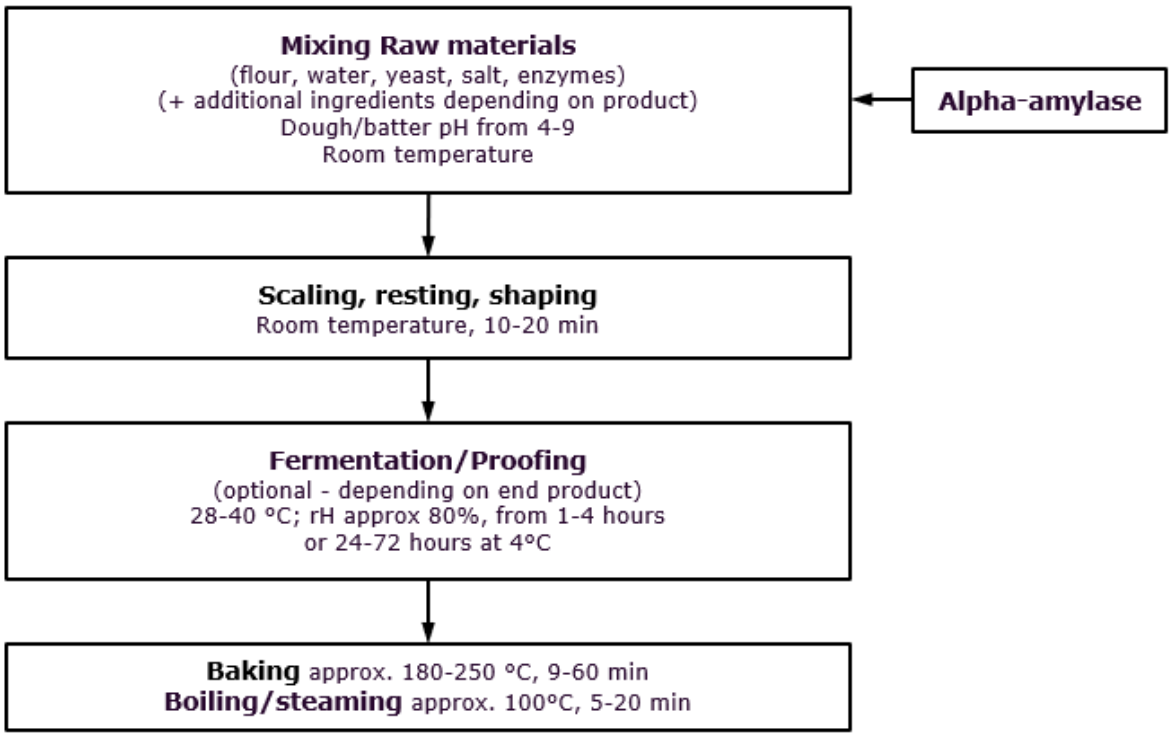
The production of baked products all includes the mixing of wheat flour and other ingredients with water in order to prepare a dough or batter, which is allowed to rest or ferment, and subsequently shaped and either baked, boiled or steamed at high temperature. The alpha-

amylase is added together with other raw materials during the dough formation step and performs its function at pH 4-9 at room temperature until the cooking step takes place.

The benefits of the action of the alpha-amylase in the production of baked products are:

- Improves dough structure and extensibility
- Helps gas production by hydrolysis of damaged starch into maltose, used by the yeast during fermentation

A typical process for production baked products:



Fate of the enzyme during food processing

In baked products production process, the alpha-amylase is denatured by heat during the baking, boiling or steaming step.

Production of cereal-based products other than baked

Alpha-amylases can also be used in the production of cereal-based products other than baked, such as pasta, breakfast cereals and snacks.

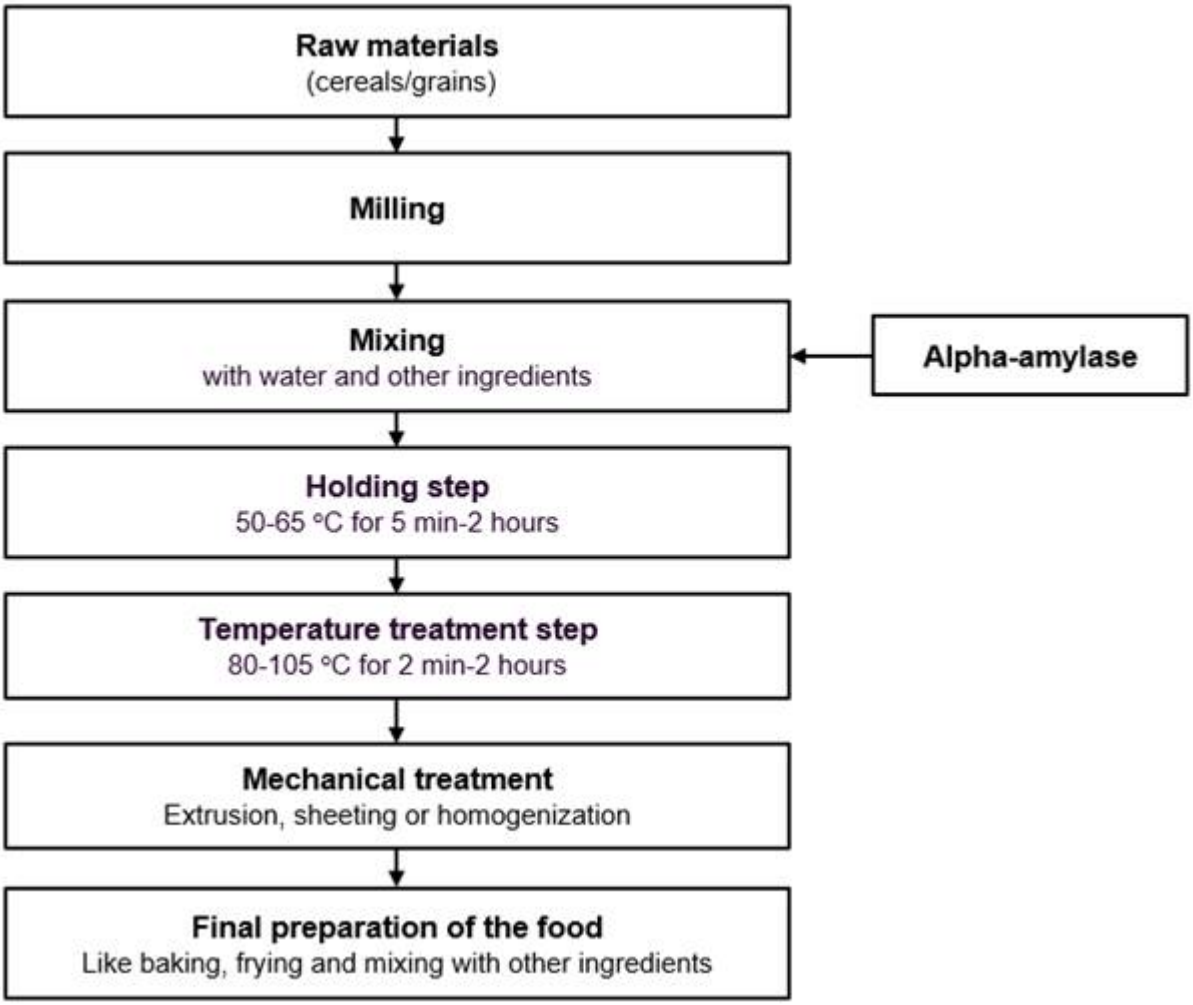
The grain based raw materials (e.g. wheat, oat, rice, rye, barley or corn) are milled, and water is added to give a slurry with 5–50 % solids, depending on the final product. The food enzyme is added to the slurry before the holding step (50-65°C at 2 min to 2 hours) in order to reduce

viscosity of the slurry/batter/dough to allow for smooth processing conditions throughout the production process. The slurry/batter/dough is heat treated before further processing like extruding or sheeting followed by a baking or frying step for breakfast cereals and snacks, or a homogenisation step for soft or semi-liquid food products.

The benefits of the action of the alpha-amylase in cereal-based processes are:

- Improved processing through viscosity reduction
- Viscosity reduction allowing for increased inclusion of cereals in food

A typical process for production of cereal-based products other than baked:



Fate of the enzyme during food processing

In production of cereal-based products other than baked, the food enzyme is denatured by heat during the temperature treatment step or in the final preparation step by baking at high temperatures or frying.

Production of plant-based analogues of milk and milk products

Alpha-amylases have been used in the production of plant-based analogues of milk and milk products for several years.

The plant based raw materials are milled, and water is added to give a slurry with 15–25 % solids, depending on the final product. The slurry is heated to facilitate the use of the enzyme followed by a temperature treatment (pasteurisation) step. The final slurry undergoes different types of processing, among others a homogenisation step for cereal-based beverages.

During the temperature treatment step, gelatinised starch contributes to a very high viscosity and thereby processing difficulties. The alpha-amylase can be used to reduce the viscosity to allow for a smooth processing of the cereal.

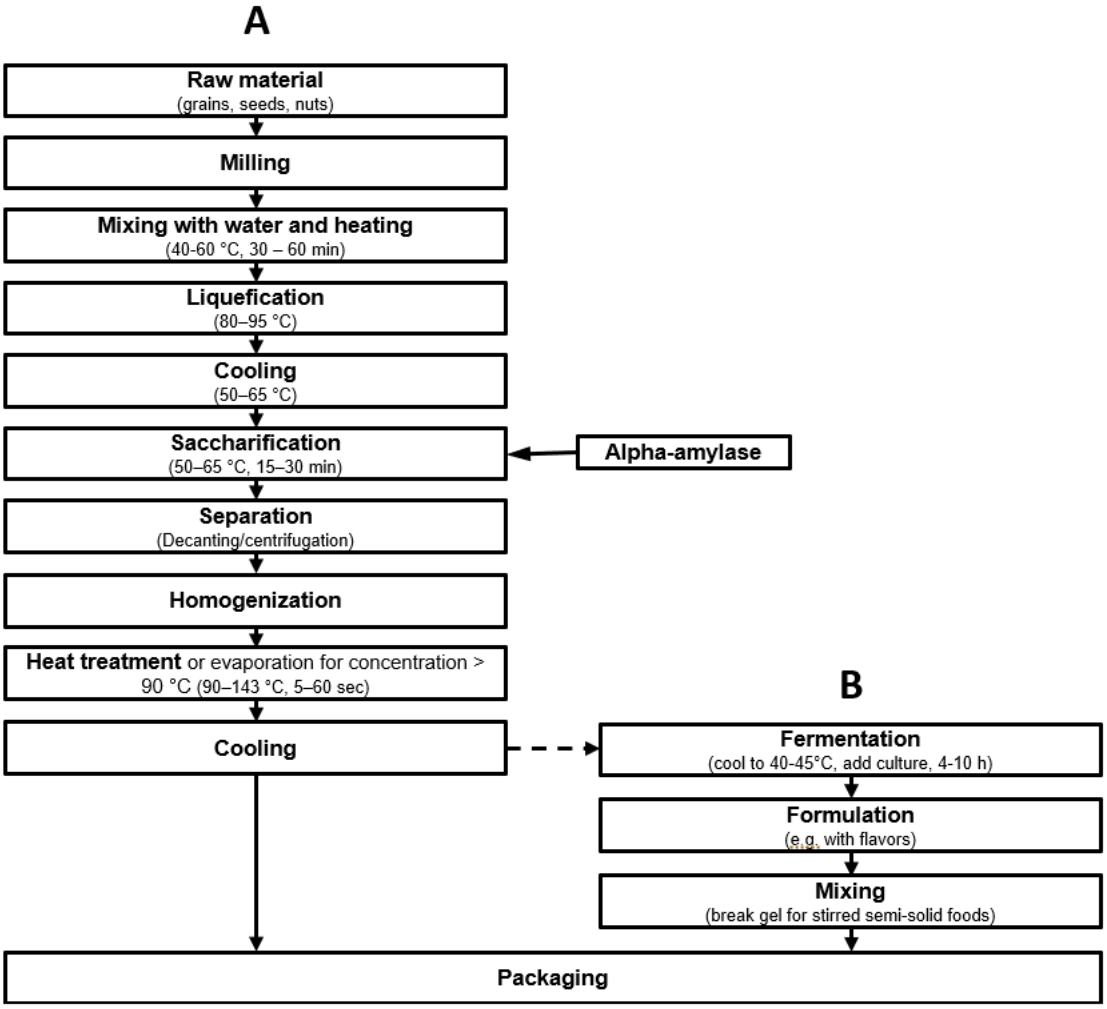
The alpha-amylase is added to the slurry during the hydrolysis step, which is then heated. This can be done in a two-step process with a 50–65 °C holding step for up to 30 minutes to enable for optimal action of the enzyme. After the hydrolysis is finished, insoluble matter is removed by decanting or centrifugation and the liquid plant-based analogue is homogenized, heat-treated and packaged (A).

The liquid dairy analogue can be further processed into plant-based yoghurt or cheese analogues (B). Culture/enzymes is added to plant based dairy analogue and undergo fermentation. Afterwards the semi-solid foods are mixed with flavours, stirred/mixed and packaged.

The benefits of the action of the alpha-amylase in production of plant-based analogues of milk and milk products are:

- Improved processing of cereals through viscosity reduction of gelatinised starch
- Viscosity reduction allowing for increased inclusion of cereals in food/beverages

A typical process for production of plant-based analogues of milk and milk products:



Fate of the enzyme during food processing:

In plant-based analogues of milk and milk products production process, the alpha-amylase is denatured by heat during the heat treatment step.

Production of starch and gluten fractions

Grains are highly complex raw materials causing technical challenges during processing where milled grain is separated into bran and fractions of starch, gluten/protein and solubles.

In the production process, grains are soaked in water to soften the kernels, milled and mixed with water. Bran is removed during this process. The slurry goes for homogenisation/agglomeration and separation into fractions of starch, gluten/protein and solubles (pentosans). Hemicellulases (such as xylanase) can be used to facilitate the separation. The separated fractions are washed, dewatered and dried. Only the starch and gluten/protein fractions are used in food processing.

The separated bran is used for animal feed.

The starch fraction is mainly further processed to syrups, such as glucose, maltose, high fructose and other syrups, which are used as ingredients in a number of food products. But it might also be used as a direct ingredient in food, e.g. in baking.

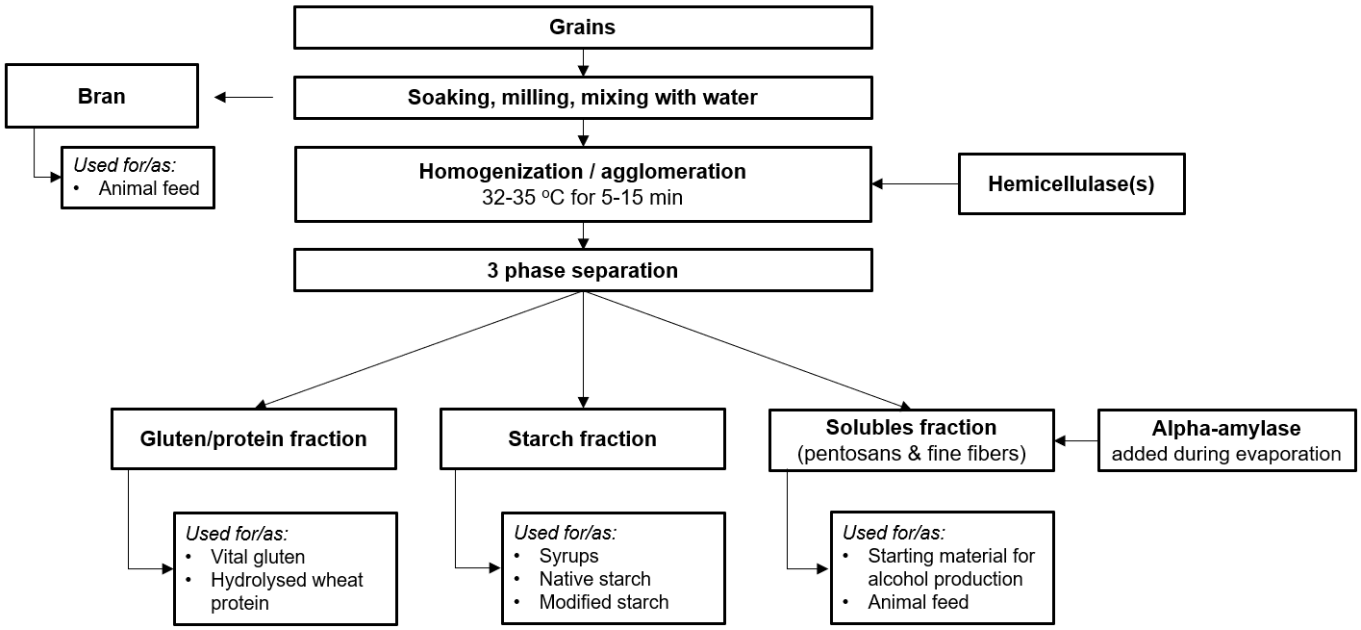
The gluten/protein fraction can also be used in baking to improve the properties of the flour or in other food applications like processing of meat.

The solubles fraction (pentosans) is used for animal feed and/or as starting material for ethanol production. Alpha-amylase can be added in the solubles fraction during evaporation to remove residual starch in this fraction, as retrogradated starch otherwise makes a strong gel and increases viscosity which has a negative impact on processability.

The benefits of the action of alpha-amylases added to the solubles fraction during evaporation are:

- Reduce viscosity
- Improve processability

A typical grain separation process is shown below:



Fate of the enzyme during food processing:

Alpha-amylase can be used during evaporation of the solubles fraction to improve processability by removing residual starch in the fraction and thereby reduce viscosity otherwise generated by starch retrogradation during the drying step. The alpha-amylase is used late in the process, i.e. after washing of the fraction, and remains in the solubles fraction. However, the solubles fraction is used only for animal feed or as starting material for alcohol production, and since the alcohol production process includes a distillation which removes all

enzyme TOS, there is no TOS in the final alcohol and consequently no dietary exposure of the alpha-amylase. In addition, the alpha-amylase object of the present dossier is also intended for use in production of distilled alcohol as described and evaluated in the original dossier.

Production of refined and unrefined sugar

Alpha-amylases can be used to remove starch during production of refined sugar.

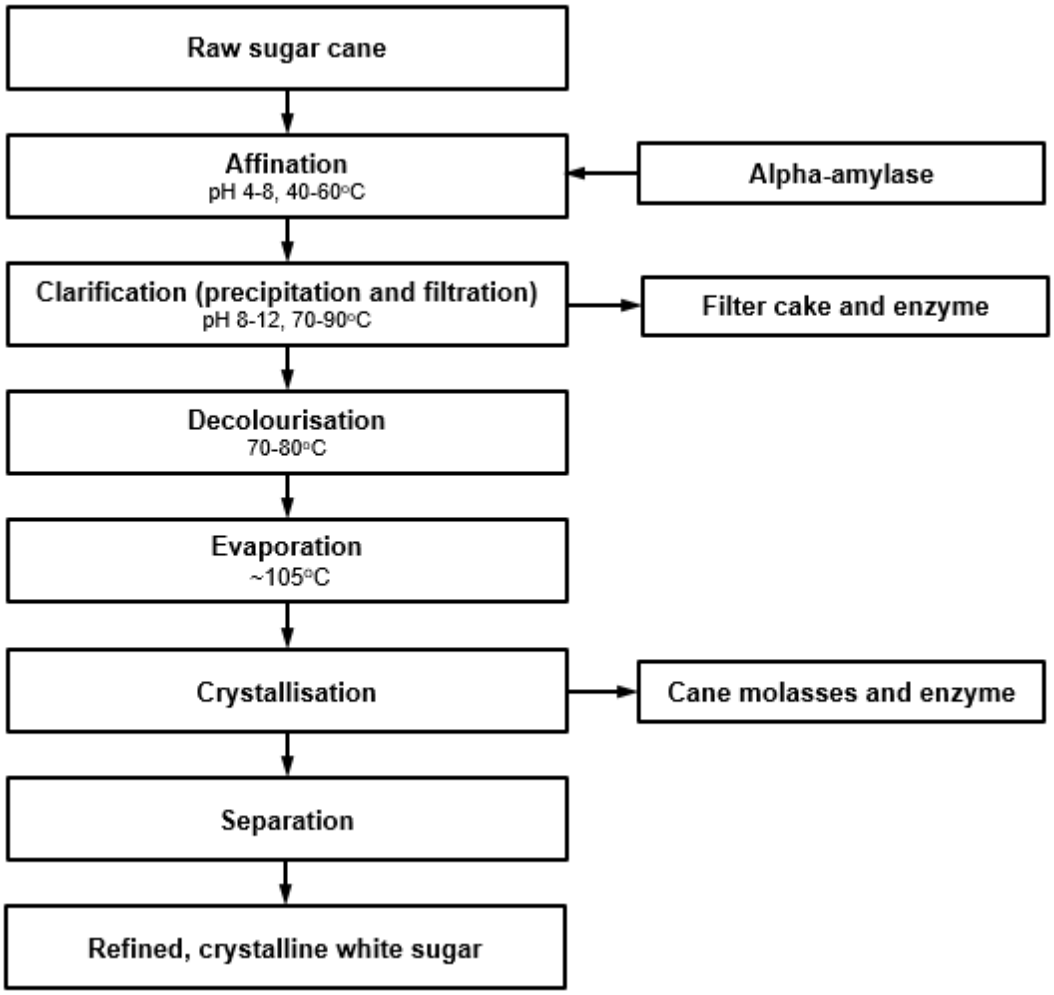
Raw materials such as sugar cane tops and leaves, and sugar beets, contain starch and starch-like components that may impair sugar crystallisation as well as allowing increased co-precipitation of starch. This will give higher starch levels in the crystalline sugar as well as lower levels of starch in the molasses fraction. The molasses fraction is primarily used for feed and in biofuels production.

During sugar refining, a high starch content in raw sugar causes filtration difficulties and an inferior crystallisation process and will reduce sugar yield. Furthermore, the co-precipitated starch may cause a visible haze when the refined sugar is dissolved in water, an unacceptable property in, e.g., the soft drinks industry. Using alpha-amylase in the process will hydrolyse starch to a level where starch does not impair the crystallisation process.

The benefits of the action of alpha-amylases are:

- Increased solubility, thus facilitating sugar crystallisation and separation
- Increased purgeability at the separation step due to reduction in viscosity
- Increased crystalline sugar yield

A typical process for refined sugar production is shown in the diagram below:



Fate of the enzyme during food processing

In refined and unrefined sugar production the food enzyme is removed during various production steps including the crystallisation step meaning that refined, crystalline white sugar will not contain any enzyme. The molasses leaving the process during processing may contain minor amounts of heat denatured enzyme, which is primarily used for feed and in biofuels production.

15. The effect of the food enzyme on the final food

Effect of *the presence* of residues of food enzymes on the final food

As explained elsewhere in this dossier enzymes are catalysts of enzymatic conversions. Therefore, in general for all food enzymes used in food processing it is not the enzyme itself, but the result of the enzymatic conversion that determines the effect in the food or food ingredient (including raw materials). This effect remains, irrespective of whether the food enzyme is still present or removed from the final food.

In general, the enzyme exerts its function during food processing. Like the endogenous enzymes naturally present in the raw materials, it does not have any function in the final food.

It should be noted that in order for a certain enzyme to be able to perform a technological function in a final food (or anywhere else), several conditions must be fulfilled at the same time:

- the enzyme must be in its native form, i.e. not denatured, AND
- the substrate for the enzyme must still be present in the food, AND
- the enzyme must be free to move (able to reach the substrate), AND
- conditions like pH, temperature and water content/availability must be favourable for the specific enzyme

If just one of these conditions is not fulfilled, the enzyme cannot work.

As described in the section '*The function and technological need, including a description of the typical process(es) in which the food enzyme may be applied*', the action of the food enzymes takes place during the food manufacturing process, and each food manufacturing process includes processing steps that will either remove or heat denature the enzyme. Consequently, any presence of residues of the food enzyme in the final food will not perform any technological function in the food.

In other words, the function of the enzyme is over long before the food product is available for delivery to consumers.

Effect on the final food of *the use* of the food enzyme during food processing

In general, use of an enzyme during food processing will always create an effect (noticeable or not), like any other process parameter used during processing, e.g., filters will remove precipitates, evaporation will remove water, chopping will mince the food. If no effect was created by a certain process parameter, there would be no reason to use it. The effect created by a certain food enzyme during food processing (noticeable or not) will remain in the food independent of whether residues of the enzymes are present in the final food or not.

As described in this dossier, the alpha-amylase is used in food processing to hydrolyse starch.

The enzyme is denatured or removed during processing, and therefore it does not exert a technological function in the final food.

The effect of the food enzyme on the final food is linked to its intended use to hydrolyse starch, and how this is translated into benefits for the various food manufacturing processes.

The use of the food enzyme in food manufacturing processes are described in the section '*The function and technological need, including a description of the typical process(es) in which the food enzyme may be applied*'. Some of these uses provide benefits that have no noticeable effects on the final food while others may have a noticeable effect on the final food. The benefits are however not exclusively obtainable by means of enzyme treatment but can be achieved without the use of enzymes through, e.g., modified production processes (maybe more expensive or less environmentally friendly), or recipe changes.

16. Information on why the use would not mislead the consumer

As described in this dossier the food enzyme performs its action during food processing, and it does not perform any technological function in the final food. The final food products or their characteristics are not different from what is expected by the consumer based on recipe and appearance.

There are no reasons to assume that the use of the food enzyme could lead to a food which would be misleading in terms of ingredients used, nutritional properties or nature or quality of the final food. The application of food enzymes does not lead to the potential use of inferior raw materials. Instead, food enzymes are used to liberate the full potentials of a raw material and to replace production processes which are less environmentally friendly. Enzymatic processes in general occur in nature and they therefore do not influence the naturalness of the production process or the final food.

Furthermore, alpha-amylases have been used in the described food manufacturing processes for many years and have been specifically approved for these processes in Denmark and France and in countries outside the EU.

Considering the above, there is no reason to believe that the use of the food enzyme in food processing could be misleading for the consume

17. List of Reference

Beg QK, Kapoor M, Mahajan L, Hoondal GS (2001) Microbial xylanases and their industrial applications: a review. Appl Microbiol Biotechnol, 56, 326-338.

EFSA CEP Panel (2021). Scientific Guidance for the submission of dossiers on Food Enzymes

Pence JW (1953) Panary Fermentation. Current Status of Problems. Journal of Agricultural and Food Chemistry, 1 (2), 157-161.

Reed G (1966) Starch, starch syrups and dextrose. Enzymes in Food Processing. Academic Press Inc., New York, 255-272.

Uhlig H (1991) Stärkeindustrie. Enzyme arbeiten für uns. Technische Enzyme und ihre Anwendung. Carl Hanser Verlag, München, 214-241.

Appendix A - Suitability checklist for food enzyme applications

The suitability check list is a document prepared by EFSA for the administrative verification of the suitability of the dossier. It can be used by applicants when building up dossiers for applications on food enzymes according to Regulation (EC) No 1331/2008 and Regulation (EU) No 234/2011. It does not substitute in any case for the requirements concerning the preparation of a dossier referred to in Regulation (EU) No 234/2011 and in EFSA's guidance documents for applicants. The checklist is based on the requirements of the EFSA Scientific Guidance for the submission of dossiers on Food Enzymes[1] (for brevity, in this checklist, this guidance is called "Scientific guidance").

This form should be submitted using a common word processing format (e.g. MS Word).

The completion of this checklist by the applicant does not replace the suitability check of the applications conducted by EFSA before they are considered valid.

Regarding the data required for risk management, these are not covered by the suitability checklist as they are not assessed by EFSA. For information on the submission of these data, the applicant should consult the European Commission Practical guidance for applicants on the submission of applications on food additives, food enzymes and food flavourings.[2]

How to fill in the suitability checklist?

This checklist allows applicants to identify which information has been provided, not provided (to be justified) and/or is not considered relevant, depending on the different parts/sections. Applicants should select the respective boxes. The definitions of the different options are detailed below:

- **Information provided**: The relevant section/subsection/paragraph is required by the relevant guidelines/guidance and the information is provided by the applicant in the relevant section/subsection/paragraph of the dossier.
- **Not provided (to be justified)**: The relevant section/subsection/paragraph is required by the guidelines but the information is not provided by the applicant in the relevant section/subsection/paragraph of the dossier. Appropriate justification for the omission of that data needs to be provided in the relevant section/subsection/paragraph of the dossier.
- **Not relevant**: The relevant section/subsection/paragraph is not relevant, either because it is not required or due to the nature or use of the substance. No specific justification needs to be provided in the dossier.

Comments can be added at the end of each part/section in the designated box.

All the fields in blue are reserved for EFSA's use.

Disclaimer: This checklist is provided by EFSA as help in the authorisation process. However, users are reminded that the text of Regulation (EC) No 1331/2008 and Regulation (EU) No 234/2011 are the only authentic legal reference and that the information in this checklist does not constitute legal advice. EFSA does not accept any liability with regard to the content of this checklist.

[1] EFSA Scientific Guidance for the submission of dossiers on Food Enzymes
[2] European Commission Practical guidance for applicants on the submission of applications on food additives, food enzymes and food flavourings

	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
Administrative data				
▪ Pre-application identification number(s) is inserted in the e-submission system, if created in Connect.EFSA	☐	☐	☒	
▪ Information on any data sharing agreement in place	☐	☐	☒	
▪ Information on any authorisation in non-EU countries	☐	☐	☒	
▪ Information on product falling within the scope of Regulation (EC) No 1829/2003 on genetically modified food and feed	☐	☐	☒	

Comments (for applicant's use):
Click or tap here to enter text.

	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
Format requirements				
▪ All files provided are electronically searchable and accessible to allow downloading and printing of the file	☒	☐	☐	
▪ Copy of bibliographic references (each reference is submitted as a separate PDF document)	☒	☐	☐	

Comments (for applicant's use):
Click or tap here to enter text.

	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
Notification of studies				
▪ Studies commissioned or carried out in support of the application are notified in EFSA's database of study notifications	☐	☐	☒	
▪ Study identification generated by EFSA's database of study notifications is included as metadata for each study submitted in the application	☐	☐	☒	
▪ In case a study is provided, but no notification has been inserted in EFSA's database of study notifications, a justification for the non-notification is provided	☐	☐	☒	
▪ In case a study has been notified in EFSA's database of study notifications but not provided in the dossier, a justification for the non-inclusion in the application is provided	☐	☐	☒	
▪ If relevant, justifications for the withdrawal of a study notification, or for delayed submission of a study notification, or for any other deviation from study notification obligations	☐	☐	☒	

Comments (for applicant's use)
Click or tap here to enter text.

	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
Identity, characterisation and specifications of the food enzyme				
▪ Accepted name as recognised by IUBMB	☒	☐	☐	
▪ Systematic name if available, common name(s) and synonyms	☒	☐	☐	
▪ IUBMB, CAS, EINECS or ELINCS (when available)	☒	☐	☐	
▪ Brief description of the catalytic activity	☒	☐	☐	
▪ JECFA Specifications	☒	☐	☐	

Comments (for applicant's use):
Click or tap here to enter text.

	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
1. Food enzyme of microbial origin				
1.1 Source of the food enzyme				
▪ Whole genome sequence (WGS) analysis - required for bacterial and yeast microorganisms and recommended for filamentous fungi and other eukaryotic microorganism	☐	☐	☒	
- Samples used for WGS analysis correspond to the production organism	☐	☐	☒	
- Protocol/method for DNA extraction	☐	☐	☒	
- Library construction method, including DNA fragmentation and selections of fragment	☐	☐	☒	
- Sequency strategy, quality control, instrumentation used and base-calling method (where applicable), including all the information requested by Section 1.1.4 of the Scientific guidance	☐	☐	☒	
- In case of de novo assembly and annotation: all the information requested by Section 1.1.5 of the Scientific guidance	☐	☐	☒	
- In case of reference-based read mapping: all the information requested by Section 1.1.6 of the Scientific guidance	☐	☐	☒	
▪ Taxonomic identity based on internationally accepted classification (genus, species and strain name or code)				
- For bacteria: identification by comparison of the WGS data with the reference genome of the type strain of the expected specie (via dDDH/ANI)	☐	☐	☒	
- For yeasts: identification by confirmation with phylogenomic analysis or by alignment to a refence genome of the same species	☐	☐	☒	
- For filamentous fungi with WGS available: confirmation with phylogenomic analysis or by alignment to a refence genome of the same specie	☐	☐	☒	
- For filamentous fungi without WGS: comparison with the 18S rRNA gene and/or ITS regions and other characteristic genes with database sequences	☐	☐	☒	
- Identification by WGS analysis using *de novo* assembly and/or read-mapping approach, including all the information requested by Section 1.1.7 of the Scientific guidance	☐	☐	☒	
- Origin of the microorganism including history and purpose of modifications performed during the development of the strain	☐	☐	☒	
▪ Certificate of deposition of the production strain, including accession number	☐	☐	☒	

	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
▪ Antimicrobial resistance (applicable to all bacteria): WGS interrogated against at least two updated databases, following the indications of Section 1.1.8. and 1.1.9 of the Scientific guidance	☐	☐	☒	
▪ Toxigenicity and pathogenicity				
- Justification if a microorganism is suitable for evaluation according to the QPS approach, following the indications of Section 1.1.10.1 of the Scientific guidance	☐	☐	☒	
- If it qualifies for QPS and it is a *Bacillus:* cytotoxicity test	☐	☐	☒	
- If it does not qualify for QPS: ○ For bacteria: WGS interrogated against updated databases, following the indications of Section 1.1.10.2. and 1.1.9 of the Scientific guidance ○ For eukaryotic microorganisms: further information on toxigenicity and virulence, as requested by Section 1.1.10.2 of the Scientific guidance	☐	☐	☒	
▪ Genetically modified microorganism (GMM)				
- General purpose and description of the traits and changes in the phenotype	☐	☐	☒	
- Description of the inserted sequence(s), deletions, base pair substitutions, frameshift mutations, including all the elements requested by Section 1.1.11.2 of the Scientific guidance	☐	☐	☒	
- Characterisation of the genetic modifications using WGS data, including all the elements requested by Section 1.1.11.3 of the Scientific guidance	☐	☐	☒	
- Characterisation of the genetic modifications without WGS data (information below to be provided as requested by Section 1.1.11.4 of the Scientific guidance) ○ Characteristics of the vector ○ Description of the genetic modification process ○ Structure of any vector/donor nucleic acid remaining in the GMM ○ Genes of concern and their absence experimentally confirmed	☐	☐	☒	

	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
1.2 Manufacturing process of the food enzyme				
▪ Production flow chart	☐	☐	☒	
▪ Detailed description of the manufacturing process	☐	☐	☒	
▪ Chemical identity, CAS number and function of all agents used in the manufacturing process	☐	☐	☒	

	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
▪ Statement confirming that the production process meets the food safety management system principles[3] and compliance with Food Hygiene Regulation[4] (if outside EU, local regulations should be specified and similarity with EU requirements confirmed)	☐	☐	☒	
▪ Description of the fermentation process with a list of actual raw materials used and their function	☐	☐	☒	
▪ Description of downstream processing techniques used, with a list of actual substances used and their function	☐	☐	☒	
▪ If applicable, any additional information for food enzyme preparations, including all the elements requested by Section 1.2.3 of the Scientific guidance	☐	☐	☒	

	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
1.3. Chemical composition, properties and purity of the food enzyme with methods of analysis				
▪ Properties of the food enzyme				
- Amino acid sequence of the enzyme under assessment, corresponding molecular mass, degree of glycosylation if relevant	☐	☐	☒	
- Modifications of the enzyme by chemical treatment, if relevant	☐	☐	☒	
- Characteristic of the protein pattern of at least 3 batches representative of those intended for commercialisation and the batch(es) used for toxicological studies, if applicable	☐	☐	☒	
- Principal enzyme activity measurements with a full description of the method used	☐	☐	☒	
- Temperature and pH range over which the enzyme remains active with optimum values determined, including a full description of the method used	☐	☐	☒	
- Thermal stability that reflects the technological role of the food enzyme, including a full description of the method used	☐	☐	☒	
▪ Chemical composition of at least 3 batches representative of those intended for commercialisation and of the batch(es) used for toxicological studies, if applicable (the same batches should be used for the analysis of all endpoints) with certificate(s) of analysis and full description of the methods used or citation of an internationally recognized method(s) - enzyme activity - concentration of total proteins/ash/water/diluent(s) if any - calculation of TOS and activity/mg TOS	☐	☐	☒	

[3] Commission Notice C/2016/4608
[4] Regulation (EC) No 852/2004

	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
▪ Purity of at least 3 batches representative of those intended for commercialisation and of the batch(es) used for toxicological studies, if applicable (the same batches should be used for the analysis of all endpoints) with certificate(s) of analysis, LoD/LoQ and full description of the methods used or citation of an internationally recognized method(s)	☐	☐	☒	
- Indication of identity and concentration of chemical impurities originated from the source (e.g. mycotoxins) and/or production process (e.g. residual amounts of antibiotics, solvents, cross-linking agents in the food enzyme, if applicable)	☐	☐	☒	
- Analysis of lead content, according to JECFA Guidelines	☐	☐	☒	
- Analysis of microbiological purity, following the protocols indicated in Section 1.3.3 of the Scientific guidance	☐	☐	☒	
- Analysis of antibacterial activity, following the protocols indicated in Section 1.3.3 of the Scientific guidance	☐	☐	☒	
▪ Demonstration of the absence of viable cells of the production strain following the protocol indicated in Section 1.3.4.1 of the Scientific guidance, with method description	☐	☐	☒	
▪ Demonstration of the absence of DNA from the production strain following the protocol detailed in Section 1.3.4.2 of the Scientific guidance, with method description	☐	☐	☒	

Comments (for applicant's use)
Click or tap here to enter text.

	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
2. Food enzyme of plant origin				
2.1 Source of the food enzyme				
▪ Identification of the source by genus and species with current accepted nomenclature	☐	☐	☒	
▪ Indication of the part(s) of the plant used to extract the food enzyme	☐	☐	☒	
▪ Description of treatments/processes applied to extract the enzyme (e.g. peeling)	☐	☐	☒	
▪ Description of other characteristics to determine the selection of the source material (e.g. degree of maturity)	☐	☐	☒	
▪ Indication of the ability of the plant to produce toxic metabolites	☐	☐	☒	
▪ Specification to control the quality of raw material	☐	☐	☒	
▪ Agents applied to the raw material (e.g. pesticides)	☐	☐	☒	
▪ Indication if the source is a genetically modified plant	☐	☐	☒	

	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
2.2 Manufacturing process of the food enzyme				
▪ Production flow chart	☐	☐	☒	
▪ Detailed description of the manufacturing process	☐	☐	☒	
▪ Chemical identity, CAS number and function of all agents used in the manufacturing process	☐	☐	☒	
▪ Statement confirming that the production process meets the food safety management system principles[3] and compliance with Food Hygiene Regulation[4] (if outside EU, local regulations should be specified and similarity with EU requirements confirmed)	☐	☐	☒	
▪ Description of the extraction process with a list of actual raw materials used and their function	☐	☐	☒	
▪ Description of downstream processing techniques used, with the list of actual materials used and their function	☐	☐	☒	
▪ If applicable, any additional information for food enzyme preparations, including all the elements requested in Section 2.2.3 of the Scientific guidance	☐	☐	☒	

	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
2.3 Chemical composition, properties and purity of the food enzyme with methods of analysis				
▪ Properties of the food enzyme				
- Amino acid sequence of the enzyme under assessment, corresponding molecular mass, degree of glycosylation if relevant	☐	☐	☒	
- Characteristic of the protein pattern of at least 3 batches representative of those intended for commercialisation and the batch(es) used for toxicological studies, if applicable	☐	☐	☒	
- Principal enzyme activity measurements with a full description of the method used	☐	☐	☒	
- Temperature and pH range over which the enzyme remains active with optimum values determined, including a full description of the method used	☐	☐	☒	
- Thermal stability that reflects the technological role of the food enzyme, including a full description of the method	☐	☐	☒	
▪ Chemical composition of at least 3 batches representative of those intended for commercialisation and of the batch(es) used for toxicological studies, if applicable (the same batches should be used for the analysis of all endpoints) with certificate(s) of analysis and full description of the methods used or citation of an internationally recognized method(s) -enzyme activity -concentration of total proteins/ash/water/diluent(s) if any - calculation of TOS and activity/mg TOS	☐	☐	☒	
▪ Purity of at least 3 batches representative of those intended for commercialisation and of the batch(es) used for toxicological studies, if applicable (the same batches should be used for the analysis of all endpoints) with certificate(s) of analysis, LoD/LoQ and full description of the methods used or citation of an internationally recognized method(s)	☐	☐	☒	
- Analysis of heavy metals, pesticides residues and mycotoxins following the indications of Section 2.3.3 of the Scientific guidance	☐	☐	☒	
- Analysis of compounds of known toxicity that might be produced by the plant source, if relevant	☐	☐	☒	
- Analysis of microbiological purity, following the protocols indicated in Section 2.3.3 of the Scientific guidance	☐	☐	☒	

Comments (for applicant's use)
Click or tap here to enter text.

	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
3. Food enzyme of animal origin				
3.1. Source of the food enzyme				
▪ Identification of the source by genus and species with current accepted nomenclature (subspecies and animal breed only if relevant)	☐	☐	☒	
▪ Indication of an animal tissue(s) used for the production of the food enzyme and how it is sourced	☐	☐	☒	
▪ Indication of history of consumption of the animal tissue(s)	☐	☐	☒	
▪ Compliance with Regulation (EU) No 2015/1162, with inspection requirements and with good hygiene practices (if outside EU, local regulations should be specified and the statement on their compatibility with the EU requirements provided)	☐	☐	☒	
▪ Description of any specification set to ensure consistency and quality of the source material	☐	☐	☒	
▪ Method(s) used to ensure the absence of risks of infectivity	☐	☐	☒	
▪ Indication if the source is a genetically modified animal	☐	☐	☒	

	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
3.2 Manufacturing process of the food enzyme				
▪ Production flow chart	☐	☐	☒	
▪ Detailed description of the manufacturing process	☐	☐	☒	
▪ Chemical identity, CAS number and function of all agents used in the manufacturing process	☐	☐	☒	
▪ Statement confirming that the production process meets the food safety management system principles[3] and compliance with Food Hygiene Regulation[4] (if outside EU, local regulations should be specified and similarity with EU requirements confirmed)	☐	☐	☒	
▪ Description of the extraction process with a list of actual raw materials used and their function	☐	☐	☒	
▪ Description of downstream processing techniques used, with the list of actual materials used and their function	☐	☐	☒	
▪ If applicable, any additional information for food enzyme preparations, including all the elements requested in Section 3.2.3 of the Scientific guidance	☐	☐	☒	

	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
3.3 Chemical composition, properties and purity of the food enzyme with methods of analysis				
▪ Properties of the food enzyme				
- Amino acid sequence and molecular mass corresponding to the enzyme derived from actual species (this information can be taken from the literature)	☐	☐	☒	
- Principal enzyme activity measurements with a full description of the method used	☐	☐	☒	
- Temperature and pH range over which the enzyme remains active with optimum values determined, including a full description of the method used	☐	☐	☒	
- Thermal stability that reflects the technological role of the food enzyme, including a full description of the method used	☐	☐	☒	
▪ Chemical composition of at least 3 batches representative of those intended for commercialisation and of the batch(es) used for toxicological studies, if applicable (the same batches should be used for the analysis of all endpoints) with certificate(s) of analysis and full description of the methods used or citation of an internationally recognized method(s) -enzyme activity -concentration of total proteins/ash/water/diluent(s) if any - calculation of TOS and activity/mg TOS	☐	☐	☒	
▪ Purity of at least 3 batches representative of those intended for commercialisation and of the batch(es) used for toxicological studies, if applicable (the same batches should be used for the analysis of all endpoints) with certificate(s) of analysis, LoD/LoQ and full description of the methods used or citation of an internationally recognized method(s)	☐	☐	☒	
- Microbiological purity, following the protocols indicated by Section 3.3.3 of the Scientific guidance	☐	☐	☒	
- Absence of Hepatitis E if the enzyme is derived from pig sources	☐	☐	☒	

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	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
4. Toxicological data				
4.1 Exemptions from toxicity testing				
▪ The food enzyme is obtained from microbial sources (GM and non-GM) which meet the requirements of the QPS approach to safety assessment, as detailed in Section 4.1a of the Scientific guidance	☐	☐	☒	
▪ The food enzyme is obtained from microbial sources and substitute toxicological data are provided. Documentation is provided demonstrating that the substitute toxicological tests meet the requirements detailed in Section 4.1b of the Scientific guidance	☐	☐	☒	
▪ The food enzyme is derived from plants and animals that are consumed by the European population and the two criteria detailed in Section 4.1c of the Scientific guidance are met	☐	☐	☒	
▪ Animal derived rennet with no safety issue raised by the manufacturing process	☐	☐	☒	
▪ No (or negligible) carry-over of the food enzyme TOS in the final food product	☐	☐	☒	

	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
4.2 Test item and dose level				
▪ The batch(es) used for toxicological studies are/is representative of the batches used for commercialisation (as judged by similar or lower activity/TOS ratio)	☐	☐	☒	
▪ Ensure that the unit in which the enzyme activity is expressed is the same as of the batches used for commercialisation. If not, conversion factor is needed.	☐	☐	☒	
▪ Certificate of analysis of the batch(es) used for toxicological studies, including chemical composition and purity, as detailed in the corresponding sections of the Scientific guidance	☐	☐	☒	
▪ Depending on the test, the units are expressed as µg TOS/plate, µg TOS/mL or mg TOS/kg bw per day and selection of concentrations/doses is justified	☐	☐	☒	

	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
4.3 Genotoxicity				
▪ Bacterial reverse mutation assay (OECD TG 471) following the indications of Section 4.3 of the Scientific guidance	☐	☐	☒	
▪ *In vitro* mammalian cell micronucleus test (OECD TG 487) following the indications of Section 4.3 of the Scientific guidance	☐	☐	☒	

	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
▪ In case of one or more positive *in vitro* tests, appropriate *in vivo* follow-up of genotoxicity studies, following the indications of Section 4.3 of the Scientific guidance	☐	☐	☒	
▪ If some of the *in vitro* and/or *in vivo* genotoxicity studies are performed in accordance with another guideline (equivalent to the OECD ones), comparability documentation is needed, as detailed in the Scientific guidance, along with a statement of GLP compliance of the laboratory conducting the study	☐	☐	☒	
4.4 Subchronic toxicity				
▪ Study performed in accordance with OECD TG 408	☐	☐	☒	
▪ If the study is performed in accordance with another guideline (equivalent to the OECD 408), comparability documentation is needed, as detailed in the Scientific guidance, along with a statement of GLP compliance of the laboratory conducting the study	☐	☐	☒	
4.5 Allergenicity				
▪ Alignment-based study using the amino acid sequence of the enzyme under assessment, following the requirements of Section 4.5(1) of the Scientific guidance	☐	☐	☒	
▪ Systematic literature search for sensitisation or elicitation reactions caused by the enzyme under assessment, following the requirements of Section 4.5(2) of the Scientific guidance	☐	☐	☒	
▪ Systematic literature search of possible allergic/sensitisation reactions caused by enzymes of the same family, following the requirements of Section 4.5(3) of the Scientific guidance	☐	☐	☒	

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	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
5. Dietary exposure assessment				
5.1 Intended use of the food enzyme				
▪ List of all food manufacturing processes where the food enzyme is intended to be used, using the definition given in the EFSA document "Food manufacturing processes and technical data used in the exposure assessment of food enzymes" - In case a suitable definition cannot be found in the EFSA document mentioned above, extra care should be given to provide information for the following four points.	☒	☐	☐	
▪ Description of the technological need, of the function of the food enzyme during manufacturing and expected benefits	☒	☐	☐	
▪ Description of each food manufacturing process aligned with the ones provided in the EFSA document "Food manufacturing processes and technical data used in the exposure assessment of food enzymes", where possible	☒	☐	☐	
▪ Flow chart - with all the information requested by Section 5.1 of the Scientific guidance - to describe each food manufacturing process	☒	☐	☐	
▪ Tabular format reporting the use levels and raw material for each food manufacturing process, as recommended by Section 5.1 of the Scientific guidance	☒	☐	☐	
5.2 Removal/absence of TOS during the food manufacturing process				
▪ For food manufacturing processes not listed in the EFSA document mentioned above as processes that remove the TOS, any claim of removal of TOS (>99%) is supported by experimental and analytical data, following the indications of Section 5.2 of the Scientific guidance	☐	☐	☒	
5.3 Food enzymes that are derived from edible parts of plants or animals				
▪ Evidence of consumption of the edible plant or animal tissue, including quantity in the EU, and source of the information	☐	☐	☒	
▪ Enzyme yield factor, reporting the amount of source materials needed to obtain 1 kg of the food enzyme.	☐	☐	☒	

	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
5.4 Calculation of exposure				
▪ Exposure estimated for each food manufacturing process for which the presence of TOS in the final product cannot be excluded	☒	☐	☐	

	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
▪ Tables reporting the estimated exposure calculated with the Food Enzyme Intake Model (FEIM[5]), where available, for each process separately, following the recommended tabular format included in Section 5.4 of the Scientific guidance	☒	☐	☐	
▪ If FEIM is not available, additional information provided as required by Section 5.4 of the Scientific guidance	☐	☐	☒	
5.5 Risk characterization				
▪ Calculation of MoE	☐	☐	☒	
▪ In case of food enzyme from edible parts of plants and animals, comparison between the exposure to the food enzyme and the exposure to a similar fraction of the source material resulting from the consumption of the plant/animal part (see also section 4.1 of this checklist on the exemptions for toxicity testing)	☐	☐	☒	

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	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
Part B. Data acquired for the safety assessment of modification to an existing authorization				
▪ Indication of the previous EFSA question number – food enzyme application for which the modification of authorization is requested	☒	☐	☐	
▪ Statement confirming that the source of the food enzyme remains the same	☒	☐	☐	
▪ Statement confirming that the manufacturing process remains unchanged, if applicable	☒	☐	☐	
▪ If new manufacturing process of the food enzyme:				
- Full description (see sections 1.2 or 2.2. or 3.2 of this checklist)	☐	☐	☒	
- If changes in the process lead to a change in TOS, the concentration of any substance of concern is needed	☐	☐	☒	
- If the substance of concern is detected in the final product, exposure assessment is needed	☐	☐	☒	
- Full set of toxicological data (see section 4 of this checklist) might be needed in the absence of published data identifying and characterizing the hazard	☐	☐	☒	
▪ If extension of intended uses of the food enzyme:				
- Each proposal for an additional food manufacturing process should be clearly indicated	☒	☐	☐	

[5] Food improvement agents applications: Tools

	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
- Intended uses provided in tabular form as indicated in section 5.1 of this checklist, highlighting what was already evaluated and what is new (both in terms of use levels and processes)	☒	☐	☐	
- Follow section 5 of this checklist, as applicable	☒	☐	☐	

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	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
List of annexes, references and checklist				
▪ List of annexes	☐	☐	☒	
▪ List of references	☒	☐	☐	
▪ Checklist (Appendix A)	☒	☐	☐	

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	PROVIDED	NOT PROVIDED	NOT RELEVANT	EFSA COMMENTS
Confidentiality				
▪ The information for which a confidentiality request is submitted, is boxed or earmarked in the confidential version of the application dossier	☒	☐	☐	
▪ The information claimed confidential is blackened out in an irreversible manner in the non-confidential version of the application dossier	☒	☐	☐	
▪ Confidentiality requests are inserted in the e-submission system	☒	☐	☐	
▪ Each request clearly identifies the information with respect to which the confidentiality request is submitted	☒	☐	☐	
▪ Each request for confidentiality is supported by an appropriate justification	☒	☐	☐	
▪ A non-confidential version is provided of each document for which a confidentiality claim is presented	☒	☐	☐	

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